

Predicting Crime Severity Fairly: Tree-Based Models and Class Imbalance in Boston

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Abstract: *This study examines crime severity prediction using spatial and temporal features under class imbalance. Five tree-based models were evaluated: baseline, class-weighted, binary for minority classes, undersampled, and SMOTE-oversampled, with interaction-based feature engineering. Baseline accuracy (~50%) was dominated by the majority class, while class weighting improved balance. Balancing techniques such as undersampling and oversampling did not provide meaningful improvements and involved trade-offs in overall performance. Models trained on 2015–2018 data generalized to 2023–2025, showing temporal stability. Results highlight the challenges of imbalance and feature limitations, suggesting that richer features could improve predictive accuracy.*

1. Introduction

Predicting crime severity is a challenging task in the field of data science, particularly when relying on limited contextual features such as spatial and temporal information. While these features are often readily available in public datasets, they may not fully capture the underlying factors that influence the seriousness of an incident. Nevertheless, understanding how far such information can go in predicting crime severity remains an important and practical research question.

A central difficulty in this problem is class imbalance. In many real-world crime datasets, one severity category tends to dominate, while others are significantly underrepresented. This imbalance can lead machine learning models to favor the majority class, resulting in high overall accuracy but poor performance on minority classes. Such behavior is problematic in applications where correctly identifying less frequent but potentially more critical cases is essential.

This study investigates the extent to which standard machine learning models can effectively predict crime severity under these constraints. In particular, it focuses on tree-based methods, including decision trees and random forests, due to their interpretability and widespread use in structured data problems. Five experimental setups are considered: (1) a baseline multiclass classification model without any

balancing techniques, (2) a multiclass model incorporating class weighting to mitigate imbalance, (3) a binary classification model restricted to minority classes to assess their separability in isolation, (4) a multiclass model trained on an undersampled dataset to evaluate the effect of reducing majority-class instances, and (5) a multiclass model trained on an oversampled dataset using SMOTE to examine the potential benefits of synthetic minority-class augmentation.

In addition to evaluating model performance under different configurations, this work also examines the temporal robustness of learned patterns. Models trained on historical data from 2015–2018 are evaluated on a more recent dataset spanning 2023–2025. This allows us to assess whether the relationships between spatial-temporal features and crime severity remain stable over time, or whether model performance degrades due to shifting patterns.

The results show that while class balancing improves performance across underrepresented categories, it introduces a tradeoff with overall accuracy. Furthermore, experiments focusing exclusively on minority classes suggest that these categories are more distinguishable when not overshadowed by the majority class. Finally, the relatively stable performance across temporally separated datasets indicates that the predictive patterns learned by the models persist over time, although the overall performance suggests inherent limitations in the available features.

This work highlights both the potential and the limitations of using spatial and temporal data for crime severity prediction, emphasizing the importance of addressing class imbalance and the need for richer feature representations in future research.

2. Related Works

A substantial portion of existing research on crime data analysis using the Boston Crime Dataset (2015–2018) focuses on exploratory data analysis and basic predictive modeling. However, several studies exhibit methodological limitations that reduce the practical validity of their findings.

One common issue is the presence of data leakage, where input features are directly or indirectly tied to the target variable. For example, some approaches attempt to predict the police district using features such as `REPORTING_AREA`, despite the fact that reporting areas are hierarchically associated with districts. This introduces a strong dependency between input and output variables, leading to artificially high performance without meaningful predictive capability.

Similarly, other studies aim to predict crime severity levels (`UCR_PART`) using features such as `OFFENSE_CODE`, which is explicitly mapped to severity categories. In such cases, models, particularly decision trees, can achieve near-perfect accuracy by exploiting this direct relationship, rather than learning generalizable patterns. As a result, these models fail to provide useful insights or predictive value in real-world scenarios.

A more constrained and methodologically sound approach is presented by Yin, Afa Michael, & Afa, 2020[1], where the authors restrict the feature set to spatial and temporal attributes, specifically latitude, longitude, and hour of occurrence. Their model

achieves an overall accuracy of approximately 51%; however, a closer examination of class-wise performance reveals a significant imbalance. The error rate for the majority class (Part Three) is relatively low (approximately 4%), while the error rates for minority classes (Part One and Part Two) exceed 88%. This indicates that the reported accuracy is largely driven by the model's tendency to predict the majority class, rather than its ability to distinguish between severity levels.

These observations highlight two critical challenges in crime severity prediction: the risk of data leakage and the impact of class imbalance on evaluation metrics. The present study builds upon prior work by explicitly addressing these issues through controlled feature selection, balanced modeling techniques, and detailed class-wise evaluation. Furthermore, it extends existing analyses by examining the temporal stability of predictive patterns across datasets from different time periods.

3. Methodology

This section presents the datasets, preprocessing procedures, feature engineering, modeling approaches, and evaluation metrics employed in this study. The preparation of the 2015–2018 and 2023–2025 Boston Crime datasets is described, including data cleaning, encoding, scaling, and splitting. Interaction features are then constructed, followed by a description of tree-based model configurations. Finally, the evaluation metrics used to assess performance are detailed, with particular attention to addressing class imbalance and evaluating temporal generalization.

3.1. Datasets

The study uses the Boston Crime Dataset covering the years 2015–2018[5] for training and testing, and an updated version of this dataset containing 2023–2025[6] for further testing, providing a temporally separated evaluation. Each record contains spatial (latitude, longitude) and temporal features (hour, day, month), as well as the target variable, crime severity(UCR_PART), categorized as Part One (most severe), Part Two, and Part Three(less severe). Both the dataset exhibits significant class imbalance, with Part Three representing the majority class.

3.2. Preprocessing

Data preprocessing involved several steps to prepare the datasets for modeling. Missing values were handled and incomplete records were filtered out to ensure data quality. Categorical variables were encoded as needed, and numerical features were scaled primarily for consistency across experiments and to facilitate model training.

3.2.1 Boston Crime Dataset 2015-2018

The Boston Crime dataset (2015–2018)[5] was obtained from Kaggle. Figure 1 summarizes the structure of the raw dataset.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 327820 entries, 0 to 327819
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   INCIDENT_NUMBER        327820 non-null  object
1   OFFENSE_CODE           327820 non-null  int64
2   OFFENSE_CODE_GROUP     327820 non-null  object
3   OFFENSE_DESCRIPTION    327820 non-null  object
4   DISTRICT               326046 non-null  object
5   REPORTING_AREA         327820 non-null  object
6   SHOOTING               1055 non-null    object
7   OCCURRED_ON_DATE       327820 non-null  object
8   YEAR                   327820 non-null  int64
9   MONTH                  327820 non-null  int64
10  DAY_OF_WEEK            327820 non-null  object
11  HOUR                   327820 non-null  int64
12  UCR_PART               327727 non-null  object
13  STREET                 316843 non-null  object
14  Lat                    307188 non-null  float64
15  Long                   307188 non-null  float64
16  Location               327820 non-null  object
dtypes: float64(2), int64(4), object(11)
memory usage: 42.5+ MB

```

Figure 1. Schema of the raw Boston Crime dataset (2015–2018)

To ensure that predictions rely only on features available in a real-world setting, a subset of relevant columns was retained, as shown in Figure 2. Columns containing post-event or non-informative data were removed.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 327820 entries, 0 to 327819
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0   UCR_PART               327727 non-null  object
1   MONTH                  327820 non-null  int64
2   HOUR                   327820 non-null  int64
3   DAY_OF_WEEK            327820 non-null  object
4   DISTRICT               326046 non-null  object
dtypes: int64(2), object(3)
memory usage: 12.5+ MB

```

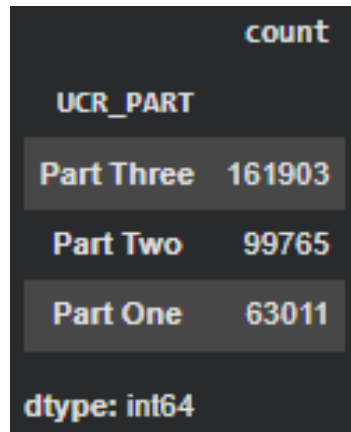
Figure 2. Schema after preprocessing and feature alignment

A new temporal feature, season, was derived from the MONTH column using the transformation:

$$season = MONTH \% 12 // 3$$

Following this step, the MONTH column was dropped to avoid redundancy. Additionally, records labeled as UCR_PART = "Other" were removed to restrict the task to the three primary severity classes.

After removing rows with missing values, the final class distribution after preprocessing is shown in Figure 3, highlighting the imbalance across severity categories.



	count
UCR_PART	
Part Three	161903
Part Two	99765
Part One	63011

dtype: int64

Figure 3. Class distribution of UCR_PART for the first dataset after preprocessing

3.2.2 Boston Crime Dataset 2023-2025

The Boston Crime dataset (2023–2025)[6] was obtained from the Boston Police Department Crime Hub, with records pre-filtered to include only incidents within this time range. Figure 4 summarizes the structure of the raw dataset.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 282480 entries, 0 to 282479
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   OBJECTID                             282480 non-null  int64
1   Incident Number                       282480 non-null  object
2   Crime                                 282480 non-null  object
3   Offense Code                          282480 non-null  object
4   Offense Description                   282480 non-null  object
5   Block Address                         246395 non-null  object
6   City                                  246427 non-null  object
7   Zip Code                             246426 non-null  object
8   BPD District                         280061 non-null  object
9   Premise Description                   246427 non-null  object
10  Weapon Description                     17708 non-null   object
11  Crime Part                            277918 non-null  object
12  Crime Category                        44073 non-null   object
13  Report Date                           282388 non-null  object
14  From Date                             280558 non-null  object
15  To Date                               92111 non-null   object
16  Hour of Day                           282480 non-null  int64
17  Day of Week                           282480 non-null  int64
18  Year                                  282480 non-null  int64
19  Quarter                              282480 non-null  int64
20  Month                                282480 non-null  int64
21  Neighborhood                           278420 non-null  object
dtypes: int64(6), object(16)
memory usage: 47.4+ MB

```

Figure 4. Schema of the raw Boston Crime dataset (2023–2025)

To ensure consistency with the 2015–2018 dataset, similar preprocessing steps were applied. Records with UCR_PART = "Other" were removed, along with entries labeled as “External” or “Outside of Boston” in the district field, as these categories were not present in the earlier dataset.

Feature selection was then performed to retain only the relevant columns required for modeling. The resulting dataset structure is shown in Figure 5. To align both datasets, column names were standardized as detailed in Table 1. Notably, the Quarter variable was renamed to season, as it represents an equivalent temporal grouping.

```

<class 'pandas.core.frame.DataFrame'>
Index: 279523 entries, 0 to 282479
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Crime Part      274990 non-null  object
1   BPD District    277146 non-null  object
2   Hour of Day     279523 non-null  int64
3   Day of Week     279523 non-null  int64
4   Quarter         279523 non-null  int64
dtypes: int64(3), object(2)
memory usage: 12.8+ MB

```

Figure 5. Schema after preprocessing and feature alignment

Table 1. Column name standardization on Boston Crime Dataset(2023-2025)

Before Renaming	Crime Part	Hour of Day	BPD District	Quarter	Day of Week
After Renaming	UCR_PART	HOUR	DISTRICT	season	DAY_OF_WEEK

After removing rows with missing values, the final class distribution is presented in Figure 6, allowing direct comparison with the preprocessed 2015–2018 dataset.

```

count
UCR_PART
Part Three  133003
Part Two    96016
Part One    43644
dtype: int64

```

Figure 6. Class distribution of UCR_PART for the second dataset after preprocessing

3.3. Encoding

Both datasets include categorical variables, namely DAY_OF_WEEK and DISTRICT. These features were encoded using ordinal mappings (e.g., Sunday = 1 through Saturday = 7 for DAY_OF_WEEK, and districts mapped in alphanumeric order). This approach was selected for simplicity and interpretability.

Table 1. Ordinal encoding for DAY_OF_WEEK

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	2	3	4	5	6	7

Table 2. Ordinal encoding for DISTRICT

A1	A7	A15	B2	B3	C6	C11	D4	D14	E5	E13	E18
1	2	3	4	5	6	7	8	9	10	11	12

Alternative encoding strategies, including label encoding and one-hot encoding, were also evaluated but resulted in negligible differences in model performance. This is consistent with the use of tree-based models, which are generally insensitive to monotonic transformations of categorical variables.

The mappings used for both variables are presented in Tables 2 and 3. Note that DAY_OF_WEEK in the 2023–2025 dataset was already provided in numeric form and therefore did not require additional encoding.

3.3. Feature Engineering

To capture interactions not explicitly represented in the raw features, two engineered features were added:

- season $\times$ hour: captures temporal patterns across seasons
- day $\times$ district: models variation by weekday and district

These interaction features were included in all experiments to evaluate their effect on model performance.

3.3. Scaling and Splitting

Prior to modeling, all features were standardized using StandardScaler from sklearn.preprocessing. Each dataset was then split into training and testing subsets using train_test_split from sklearn.model_selection with a 25% test size and random_state = 42 for reproducibility.

- 2015–2018 dataset: split into X_train, X_test, Y_train, and Y_test for training and evaluation
- 2023–2025 dataset: split similarly; only the testing portion was used for model evaluation

3.4. Models

Five experimental setups were conducted using tree-based models to evaluate the effects of class imbalance and feature engineering on crime severity prediction. Each experiment uses a Random Forest classifier with the same base parameters unless otherwise noted.

Experiment 1 – Baseline Multiclass Model

The baseline Random Forest was trained on all three severity classes without class balancing, serving as a reproduction of prior work(Yin *et al.*, 2020)[1] and establishing a reference point for comparison. The model parameters were determined using grid search with the feature set selected earlier in this study. Model parameters:

n_estimators = 200, min_samples_leaf = 2, min_samples_split=5, max_features = "sqrt", max_depth = 10, class_weight = None.

Experiment 2 – Class-Weighted Multiclass Model

The same Random Forest setup as Experiment 1, but with class_weight = "balanced". This adjustment aims to improve performance on minority classes while maintaining reasonable overall accuracy.

Experiment 3 – Minority-Class Binary Model

The Random Forest was trained exclusively on Part One and Part Two incidents, with all Part Three rows removed. This experiment evaluates the separability of minority classes when the majority class is excluded. The configuration is identical to Experiment 2.

Experiment 4 – Undersampled Multiclass Model

The baseline Random Forest configuration from Experiment 1 was applied to a dataset balanced through undersampling. The majority classes were randomly reduced using resample from sklearn.utils (replace = False, random_state = 42) until all classes matched the size of the original smallest class (Part One). This approach evaluates the impact of removing excess majority-class observations on model performance. Model parameters remain identical to Experiment 1.

Experiment 5 – Oversampled Multiclass Model (SMOTE)

The baseline Random Forest configuration from Experiment 1 was trained on a dataset balanced via synthetic oversampling. The Synthetic Minority Over-sampling Technique (SMOTE) was applied with sampling_strategy = "auto" and random_state = 42 to generate new minority-class samples until all classes were equally represented. This experiment assesses whether augmenting minority classes improves classification performance without discarding existing data. Model parameters remain identical to Experiment 1.

In addition, two multiclass Decision Tree models were trained for comparison to illustrate the limitations of simpler models under identical feature and class conditions. Both used min_samples_leaf = 3, max_depth = 10, and random_state = 42; the second model applied class_weight = "balanced".

3.4. Evaluation Metrics

Model performance was assessed using the following metrics:

- Accuracy: overall proportion of correctly classified instances, computed using accuracy_score from sklearn.metrics.
- Precision, Recall, and F1-score: calculated per class and macro-averaged to account for class imbalance, using classification_report from sklearn.metrics.
- Class-Specific Error Rate: to visualize the distribution of misclassifications across classes, derived from the confusion matrix as:

$$\text{Error per class} = 1 - \frac{\text{correct predictions for that class}}{\text{total instances of class}}$$

To evaluate temporal generalization, all models trained on the 2015–2018 dataset were tested on both the 2015–2018 and 2023–2025 datasets.

4. Results

This section presents the outcomes of the experiments conducted on the Boston Crime datasets. Performance is evaluated across multiple modeling approaches, highlighting the effects of class imbalance, feature engineering, and temporal generalization. Metrics are reported for both the 2015–2018 and 2023–2025 datasets, with a focus on overall accuracy, class-specific performance, and the effectiveness of balancing strategies.

4.1. Experiment 1 – Baseline Multiclass Model

The baseline Random Forest model, trained on all three severity classes without class balancing, achieved an overall accuracy of approximately 50% across both evaluation periods. However, this performance is misleading due to strong class imbalance.

	Part One	Part Two	Part Three
Part One	152	340	15297
Part Two	110	535	24369
Part Three	162	535	39670

Figure 1. Confusion Matrix for Experiment 1 with 2015-2018 test split

As shown in Figure 1, the confusion matrix indicates that predictions are overwhelmingly concentrated in the majority class (Part Three), with minimal correct classification of Part One and Part Two instances. This behavior is consistent with prior observations by Yin et al., 2020[1].

Table 3. Precision, recall, f1 macro averages and accuracy results from Experiment 1 with both datasets test splits

Dataset	Accuracy	F1-macro	Precision-macro	Recall-macro
2015-2018	49.71%	24.06%	41.26%	33.79%
2023-2025	48.84%	23.51%	39.08%	33.61%

Macro-averaged metrics further highlight this limitation (Table 3). Despite moderate accuracy, the F1-macro score remains low ($\approx 24\%$), with similarly reduced macro precision and recall, indicating poor performance across classes when treated equally.

Class-specific error rates (Table 4) reinforce this pattern. Error rates exceed 97% for both Part One and Part Two across both dataset splits, while remaining below 2% for Part Three. These results are consistent between the 2015–2018 and 2023–2025 datasets.

Table 4. Error rate per class derived from confusion matrix from Experiment 1 with both datasets test splits

Dataset	Error Part One	Error Part Two	Error Part Three
2015-2018	99.03%	97.86%	1.72%
2023-2025	99.30%	98.03%	1.80%

Overall, the baseline model demonstrates a strong bias toward the majority class, limiting its effectiveness for minority class prediction.

4.2. Experiment 2 – Class-Weighted Multiclass Model

Introducing class weights to address class imbalance resulted in a more balanced model across all severity classes. While overall accuracy decreased to approximately 40%, macro-averaged performance improved substantially, with F1-macro increasing to around 38% across both evaluation periods (Table 5).

Table 5. Precision, recall, f1 macro averages and accuracy results from Experiment 2 with both datasets test splits

Dataset	Accuracy	F1-macro	Precision-macro	Recall-macro
2015-2018	40.51%	38.78%	39.13%	39.77%
2023-2025	38.32%	36.26%	36.71%	37.64%

The confusion matrix (Figure 2) reflects a clear shift in model behavior compared to the baseline. Predictions are more evenly distributed across all classes, rather than being dominated by Part Three. Notably, for each class, the highest number of predictions corresponds to the correct class label. Although misclassifications remain substantial, this indicates that the model is now able to meaningfully distinguish between all three severity levels.

	Part One	Part Two	Part Three
Part One	5751	5048	4990
Part Two	6285	10312	8417
Part Three	9643	13901	16823

Figure 2. Confusion Matrix for Experiment 2 with 2015-2018 test split

This improvement is further supported by the class-specific error rates shown in Table 6. Error rates for Part One and Part Two decrease significantly compared to the baseline, while the error rate for Part Three increases, indicating a more balanced distribution of errors across classes. This pattern is consistent across both dataset splits.

Table 6. Error rate per class derived from confusion matrix from Experiment 2 with both datasets test splits

Dataset	Error Part One	Error Part Two	Error Part Three
2015-2018	63.57%	58.77%	58.32%

2023-2025	64.02%	63.31%	59.71%
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Overall, the class-weighted model demonstrates a clear tradeoff: reduced overall accuracy in exchange for substantially improved balance across classes.

4.3. Experiment 3 – Minority-Class Binary Model

To further investigate minority class performance, a binary Random Forest model was trained using only Part One and Part Two incidents. Under this setting, overall accuracy increased to approximately 56–58%, with macro-averaged F1-scores exceeding 50% across both evaluation periods (Table 7).

Table 7. Precision, recall, f1 macro averages and accuracy results from Experiment 3 with both datasets test splits

Dataset	Accuracy	F1-macro	Precision-macro	Recall-macro
2015-2018	57.94%	56.16%	56.13%	56.27%
2023-2025	56.29%	52.91%	53.34%	53.78%

The confusion matrix (Figure 3) indicates improved separability between the two classes compared to previous experiments. For Part Two, correct predictions exceed misclassifications, suggesting that the model is able to identify this class with reasonable consistency. For Part One, however, misclassifications into Part Two remain slightly higher than correct predictions, indicating that the model still struggles to fully distinguish between the two categories.

	Part One	Part Two
Part One	5224	5725
Part Two	9520	14446

Figure 3. Confusion Matrix for Experiment 3 with 2015-2018 test split

Class-specific error rates (Table 8) reflect this asymmetry. While error rates remain non-negligible for both classes, Part Two shows a lower error rate ($\approx 36\text{--}39\%$) compared to Part One ($\approx 51\text{--}53\%$), confirming that Part Two is more reliably predicted. These patterns are consistent across both dataset splits.

Table 8. Error rate per class derived from confusion matrix from Experiment 3 with both datasets test splits

Dataset	Error Part One	Error Part Two
2015-2018	51.10%	36.34%
2023-2025	52.93%	39.49%

Overall, removing the majority class leads to a substantial improvement in minority class performance. This suggests that class imbalance plays a significant role in limiting model effectiveness, although some degree of class overlap or limited feature separability remains.

4.3. Experiment 4 – Undersampled Multiclass Model

To assess the impact of undersampling on multiclass performance, the Random Forest was trained on a balanced dataset where all classes matched the size of the original smallest class (Part One). Overall accuracy remained low, failing to exceed 40% across both evaluation periods (Table 9). Macro-averaged F1-scores, precision, and recall were also comparable to Experiment 2, indicating that undersampling did not provide a meaningful improvement in model performance.

Table 9. Precision, recall, f1 macro averages and accuracy results from Experiment 4 with both datasets test splits

Dataset	Accuracy	F1-macro	Precision-macro	Recall-macro
2015-2018	39.50%	39.49%	39.68%	39.52%
2023-2025	37.31%	37.34%	37.48%	37.31%

The confusion matrix (Figure 4) exhibits patterns almost identical to Experiment 2, with the majority class still dominating misclassifications. Class-specific error rates (Table 10) are balanced across all classes, reflecting the equalized training set but confirming that the model is unable to achieve higher predictive accuracy under this sampling strategy.

	Part One	Part Two	Part Three
Part One	4126	3752	3141
Part Two	3293	4133	3381
Part Three	2808	4146	3953

Figure 4. Confusion Matrix for Experiment 4 with 2015–2018 test split

Table 10. Error rate per class derived from confusion matrix from Experiment 4 with both datasets test splits

Dataset	Error Part One	Error Part Two	Error Part Three
2015-2018	62.33%	56.86%	62.27%
2023-2025	62.56%	61.76%	63.76%

In summary, undersampling the dataset to balance class sizes did not meaningfully improve model performance. While error rates became more uniform across classes, overall predictive capability remained limited, suggesting that simply reducing majority-class instances is insufficient to overcome the challenges posed by the dataset.

4.3. Experiment 5 – Undersampled Multiclass Model

To investigate whether synthetic oversampling could improve multiclass performance, the Random Forest was trained on a dataset balanced using SMOTE, generating new samples for minority classes until all classes were equally represented. Despite this augmentation, overall accuracy remained low, slightly below Experiment 2, and macro-averaged F1-scores, precision, and recall showed no meaningful improvement (Table 11).

Table 11. Precision, recall, f1 macro averages and accuracy results from Experiment 5 with both datasets test splits

Dataset	Accuracy	F1-macro	Precision-macro	Recall-macro
2015-2018	41.11%	41.14%	41.25%	41.11%
2023-2025	37.80%	37.77%	37.93%	37.80%

The confusion matrix (Figure 5) closely resembles those of Experiments 1 and 2, with the majority class still dominating misclassifications and minority-class predictions remaining unreliable. Class-specific error rates (Table 12) are largely unchanged from previous experiments, reflecting that synthetic oversampling did not enhance separability among the classes.

	Part One	Part Two	Part Three
Part One	11830	10585	10733
Part Two	9149	11757	12440
Part Three	7988	11152	14119

Figure 5. Confusion Matrix for Experiment 5 with 2015–2018 test split

Table 12. Error rate per class derived from confusion matrix from Experiment 5 with both datasets test splits

Dataset	Error Part One	Error Part Two	Error Part Three
2015-2018	59.31%	57.47%	59.89%
2023-2025	64.31%	64.74%	57.55%

Overall, oversampling via SMOTE did not lead to meaningful improvements. These results make it clear that class imbalance is not the sole limitation: the predictive quality of the features themselves constrains model performance, highlighting the need for more informative or discriminative features to achieve higher accuracy.

4.4. Overall Discussion

The experiments demonstrate that class imbalance is the primary challenge in predicting crime severity using spatial and temporal features. The baseline Random Forest achieved moderate overall accuracy, but predictions were dominated by the majority class, with minority classes largely misclassified. Introducing class weights improved balance across classes, increasing macro F1-scores while slightly reducing overall accuracy, and binary modeling of the minority classes confirmed that these categories are distinguishable when the majority class is removed.

Experiments 4 and 5 evaluated alternative approaches to handling class imbalance: undersampling the majority classes and oversampling the minority classes using SMOTE. Both strategies failed to produce meaningful improvements. The undersampled model (Experiment 4) resulted in balanced error rates across classes, but overall accuracy remained low, under 40%. Similarly, the SMOTE-augmented model (Experiment 5) achieved slightly lower accuracy than the baseline, with macro-level performance metrics largely unchanged. These findings indicate that simply equalizing class distributions is insufficient; the limitations in predictive quality are driven largely by the available features rather than imbalance alone.

Table 9. Results from all experiments

Model	Accuracy	F1 macro	Precision macro	Recall macro	Part 1 Error	Part 2 Error	Part 3 Error
Experiment 1	49.71%	23.45%	40.41%	33.62%	99.57%	98.34%	1.24%
Experiment 2	40.47%	38.81%	39.20%	39.85%	63.47%	58.08%	58.89%
Experiment 3	58.17%	56.40%	56.37%	56.50%	50.84%	36.15%	---
Experiment 4	39.50%	39.49%	39.68%	39.52%	62.33%	56.86%	62.27%
Experiment 5	41.11%	41.14%	41.25%	41.11%	59.31%	57.47%	59.89%
Decision tree imbalanced	49.47%	25.88%	39.27%	34.13%	97.94%	95.58%	4.06%
Decision tree balanced	39.32%	38.11%	38.85%	39.59%	60.30%	59.11%	61.78%

Interaction-based feature engineering, including combinations such as season $\times$ hour and day $\times$ district, produced little improvements. Across both the 2015–2018 and 2023–2025 datasets, performance patterns remained consistent, suggesting that spatial-temporal relationships in crime severity are relatively stable over time.

Additional analyses using decision trees, both imbalanced and class-weighted, reinforced these findings. Balanced trees reduced majority class dominance and yielded similar macro-level improvements as the class-weighted Random Forest, demonstrating that the observed patterns are robust across model types. Overall, class-weighted models offer the best compromise between fairness and accuracy. Temporal tests indicate that the learned patterns are meaningful, while the results from Experiments 4 and 5 highlight the inherent limitations of the available features for fully resolving minority class predictions.

5. Conclusion

This study evaluated tree-based models for predicting crime severity using spatial and temporal features, highlighting the challenges of class imbalance and limited feature information. Baseline models achieved moderate accuracy but largely ignored minority classes, while class-weighted Random Forests improved fairness across all categories. Binary modeling of minority classes confirmed that these categories are separable when the majority class is removed. Additional experiments with undersampling and SMOTE-based oversampling showed that simply balancing class distributions does not substantially improve performance, emphasizing that predictive limitations are driven more by feature quality than imbalance alone. Feature engineering with interaction terms provided minor gains, and models trained on 2015–2018 data generalized well to 2023–2025, demonstrating temporal stability.

Overall, the results indicate that while spatial-temporal features alone are insufficient for highly accurate crime severity prediction, careful handling of imbalance, thoughtful feature design, and richer feature sets can enhance performance. Future work could explore incorporating new features available in the more recent dataset to improve predictive accuracy and better capture the factors that drive crime severity.

6. References

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